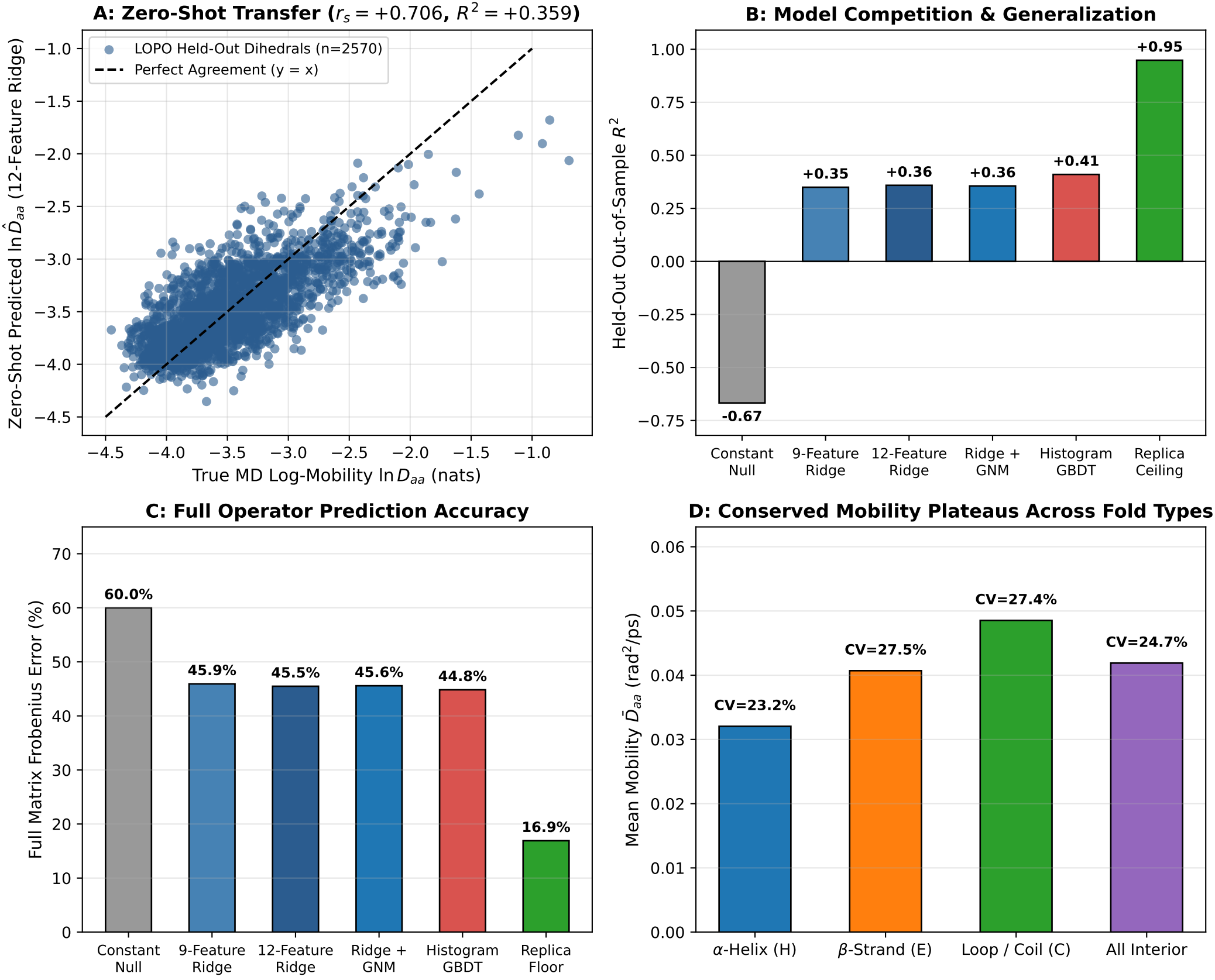

A: Zero-Shot Transfer ($r_s = +0.706$, $R^2 = +0.359$)
LOPO Held-Out Dihedrals (n=2570)
Perfect Agreement (y = x)
Zero-Shot Predicted $\ln \hat{D}_{aa}$ (12-Feature Ridge)
True MD Log-Mobility $\ln D_{aa}$ (nats)
B: Model Competition & Generalization
Held-Out Out-of-Sample R^2
-0.67
+0.35
+0.36
+0.36
+0.41
+0.95
Constant Null
9-Feature Ridge
12-Feature Ridge
Ridge + GNM
Histogram GBDT
Replica Ceiling
C: Full Operator Prediction Accuracy
Full Matrix Frobenius Error (%)
60.0%
45.9%
45.5%
45.6%
44.8%
16.9%
Constant Null
9-Feature Ridge
12-Feature Ridge
Ridge + GNM
Histogram GBDT
Replica Floor
D: Conserved Mobility Plateaus Across Fold Types
Mean Mobility $\bar{D}_{aa}$ (rad^2/ps)
CV=23.2%
CV=27.5%
CV=27.4%
CV=24.7%
α-Helix (H)
β-Strand (E)
Loop / Coil (C)
All Interior